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INFORMATION SYSTEM ANALYSIS AND DESIGN

Information System Investigation Report: Requirements Identification and Analysis

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Chapter 1

REQUIREMENTS IDENTIFICATION

1.1 Introduction to the Problem

1.1.1 Business context

Fashion e-commerce differs from the sale of a single, undifferentiated item. One product design can be offered in several colors and sizes; every combination must be identified, priced, displayed, counted, picked, and recorded consistently. In the ABC Fashion case described in the business investigation, variant information is fragmented, inventory updates rely on spreadsheets, and size information does not sufficiently support customers' choices. These conditions contribute to canceled orders, incorrect fulfillment, and size exchanges.

This chapter identifies the business problems, stakeholders, needs, system boundary, and requirements. The case is an academic analysis; no new field interviews, measured outcomes, or operational statistics are claimed.

1.1.2 Organization and business model

ABC Fashion is a medium-sized direct-to-consumer online fashion retailer in this case. It works with garment suppliers, a payment provider, and a shipping provider. Seasonal collections and unsold stock influence purchasing and reporting. The proposed information system supports sales and internal coordination but does not itself manufacture goods, move parcels, or settle payments.

1.1.3 Products and target customers

The merchandise includes clothing, dresses, and accessories. A product design has a product identifier; a specific color-size combination has its own stock-keeping unit (SKU). A white shirt in size M and a black shirt in size L therefore represent different variants for inventory and order processing. Target customers are online shoppers who depend on reliable photographs, size charts, product availability, and an accessible exchange process. Internal users need the same variant identity to handle orders and stock accurately.

1.2 Review of Current Business Operations

1.2.1 Main activities and existing workflow

The business investigation describes collection and variant management, customer information, search and product viewing, size support, carts, orders, payment, picking, delivery, exchanges, reviews, and management reporting. The current workflow moves from product information to customer selection, an availability check, order processing, picking, shipment, and after-sales support. Inventory is still handled with general product codes and manual consolidation. Reserving inventory by SKU and generating SKU-level pick lists are proposed capabilities; they must not be mistaken for features of the current operation.

1.2.2 Participating departments and external parties

Administrators maintain catalog data. Merchandising staff plan collections; sales staff handle orders; warehouse staff count, pick, pack, and receive returned items. Customer service addresses questions and exchanges. Marketing uses product and sales information, while managers evaluate performance. Customers, garment suppliers, the payment provider, and the shipping provider participate from outside the organization. A stakeholder need not be a direct software actor: this distinction becomes important in Chapter 2.

1.2.3 Ten major business activities

The business investigation distinguishes ten activities. Their inclusion describes the enterprise workflow; it does not imply that each activity already has a fully specified software function.

Activity	Business work and information
Collection and variant management	Maintain categories, fabric specifications, size charts, color/size SKU matrices, and photographs for each color.
Customer and measurement profiles	Keep contact details, addresses, purchase history, height, weight, and measurements for possible sizing assistance.
Browsing and size selection	Search and filter by available variants; inspect lookbooks and size information.
Variant shopping cart	Store the exact SKU and quantity chosen and check availability at checkout.
Fashion order processing	Record orders, reserve variants under an agreed stock rule, and create SKU-level pick lists.
Payment management	Request online payment authorization or record a COD method; reconcile its actual status.
Matrix inventory management	Track color-size stock and identify unavailable or slow-moving sizes.
Shipping and returns	Issue waybills, follow delivery, and coordinate reverse logistics for exchanges.
Visual reviews	Collect ratings, fit feedback, fabric comments, and customer photographs where permitted.
Fashion analytics	Report sales by collection and variant and returns by design to guide future purchasing.

Table 1.1: The ten business activities and their principal information needs.

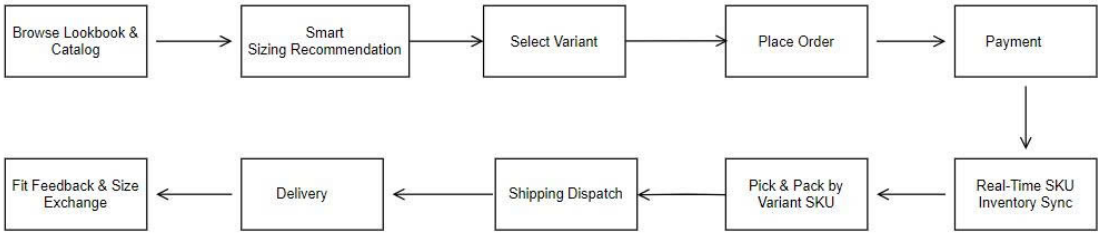


Figure 1.1: High-level business process from browsing and size selection through payment, SKU fulfillment, delivery, and fit feedback.

The process figure includes a smart sizing step that the business investigation proposes. Its presence in a preliminary business process does not, by itself, supply a testable sizing FR in the committed FR01–FR20 catalogue. This distinction is maintained throughout both chapters.

1.2.4 Business information

Inputs include product and variant data, size charts, customer measurements, SKU selection, quantities, addresses, payment responses, shipment updates, and exchange requests. Outputs include order confirmations, pick lists, waybills, status notifications, and reports. Persistent information covers customers, catalog entries, orders and lines, payments, stock records, shipments, returns, and reviews. The same item of information may be produced by one activity, used by another, and stored for later reconciliation.

Information	**Source**	**Main use**
Product, color, size, SKU	Administration, merchandising	Variant selection and correct warehouse picking.
Size chart, measurements	Business, customer	Size comparison and potential advice.
Order and order lines	Customer, system	Sales, warehouse, and customer-service processing.
Quantity by SKU	Warehouse, order operations	Availability checks and prevention of overselling.
Payment, waybill	External services	Transaction and shipment reconciliation.
Exchange, review	Customer, warehouse	After-sales work and product feedback.
Business report	Transaction summaries	Sales and collection decisions.

Table 1.2: Business information summarized from the business investigation.

1.3 Stakeholders and Their Needs

1.3.1 Internal and external stakeholders

Internal stakeholders are administrators, merchandising staff, sales staff, warehouse staff, customer-service staff, marketing staff, and managers. External stakeholders are customers, garment suppliers, the payment provider, and the shipping provider. Administrators need consistent product and SKU definitions. Warehouse staff need correct quantities and picking instructions. Managers and merchandisers need timely sales, stock, and return information. Customers need to choose the right variant and track their transactions.

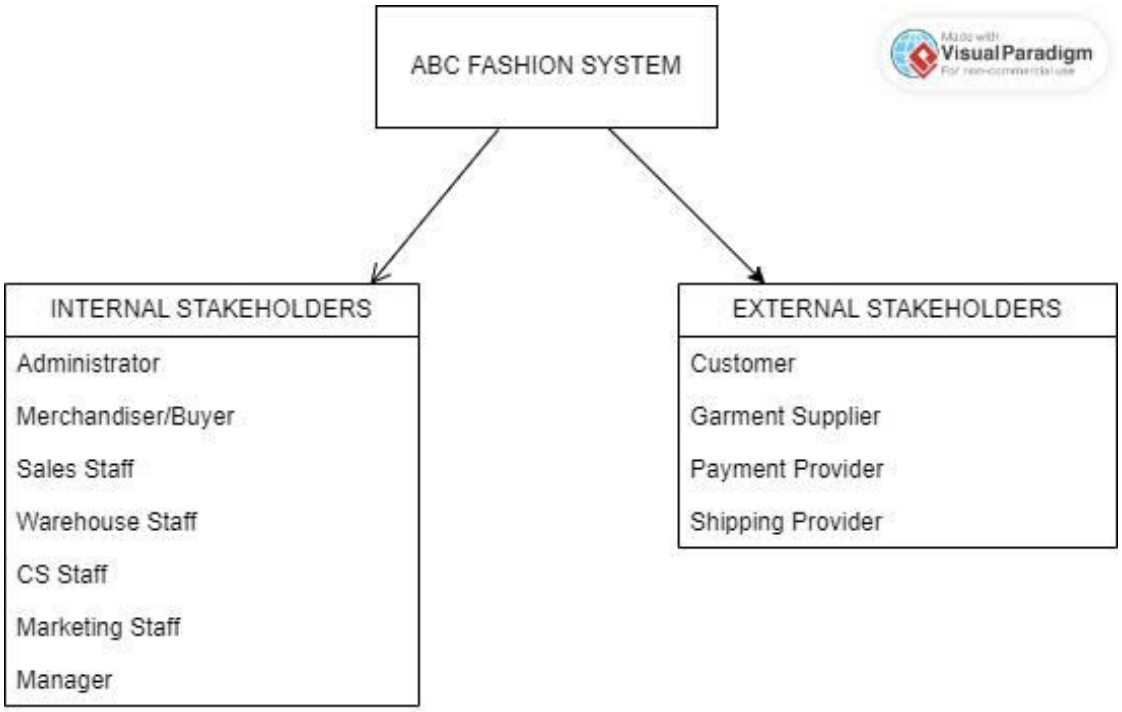


Figure 1.2: Stakeholder map distinguishing seven internal roles and four external parties.

1.3.2 Roles and interests of the stakeholder groups

Stakeholder	**Principal interest**
Customer	Reliable fit and size information, accurate delivery, and manageable returns.
Administrator	A consistent, rapidly changing catalogue and valid product variants.
Merchandiser / Buyer	Sales and stock data for collections and size distributions.
Sales Staff	Correct order, customer, and sizing information.
Warehouse Staff	Physical and available quantities for each color-size SKU.
Customer Service Staff	Fast access to order history, fit feedback, and return state.
Marketing Staff	Sales trends relevant to campaigns and promotions.
Manager	Timely sales, stock, and return analysis.
Garment Supplier	Demand and potential reorder information by style and size.
Payment Provider	Correct and secure transaction data.
Shipping Provider	Precise addresses, waybills, and return pickup details.

Table 1.3: Distinct interests of the eleven stakeholder groups.

1.3.3 Stakeholder needs

The following identifiers organize ten needs already represented in the requirements specification; they do not assert that new interviews occurred.

ID	Stakeholder	Need	Expected benefit
NC01	Customer	Search by text, voice, or image.	Find relevant products quickly.
NC02	Customer	View variants and add the desired one to the cart.	Know color, size, price, and availability.
NC03	Customer	Follow order and shipment status.	Understand purchase progress.
NC04	Administrator	Manage catalog and variants.	Consistent product and SKU data.
NC05	Manager	View dashboards and reports.	Monitor sales, inventory, and returns.
NC06	Warehouse staff	View pick lists and update stock by SKU.	Pick the correct variant and avoid overselling.
NC07	Warehouse staff	Receive returns and adjust stock.	Distinguish resalable and damaged items.
NC08	Customer service	Access customer feedback.	Support complaints and reviews.
NC09	Payment provider	Verify transactions and process refunds.	Reconcile payment state.
NC10	Shipping provider	Create waybills and update delivery/return state.	Track shipping and exchanges.

Table 1.4: Stakeholder needs used for traceability.

NC08 is only partly covered by FR18, which allows customer comments but does not specify a customer-service complaint workflow. NC10 is reflected in FR14–FR15, although exchange formats, timing, and error handling remain to be clarified. Accurate stock benefits both the warehouse and customers even though their immediate tasks differ.

1.4 Information Analysis

1.4.1 Information produced

The business produces order confirmations that identify color and size, pick lists by SKU and shelf location, electronic waybills, VAT invoices, sales reports by collection, return reports by size, and delivery notifications. These are proposed or existing business information outputs, depending on the current workflow; the presence of an output here is not proof that an automated module already exists.

1.4.2 Information consumed

Business activities consume merchandising size specifications and lookbook images, customer body measurements, SKU-level stock, payment verification, shipping status messages, and fit feedback. Order validation depends on current stock; shipping status depends on information from the external provider.

1.4.3 Information stored

The shared information base retains customer profiles and measurements; the collection tree; product style, color, size, SKU, and listed price; variant order lines; payment transaction history; warehouse shelf and stock records; waybills and exchanges; and reviews with any permitted customer images.

1.4.4 Information catalogue

Information	Producer	Consumer	Purpose
Product variant data	Administrator	Customer, sales	Select and order the correct color/size.
Measurement profile	Customer	System, sales	Support sizing advice when this capability is specified.
Variant order data	Customer, system	Warehouse, sales	Pack the exact SKU and quantity.
Matrix stock data	Warehouse, system	Customer, manager	Identify availability and prevent overselling.
Sales and trend reports	System	Manager, buyer	Plan collections and size quantities.
Fit feedback	Customer	Manager, shoppers	Describe actual form and fit.

Table 1.5: Information catalogue connecting producers, consumers, and purposes.

1.5 Problems in the Current Situation

1.5.1 Problems, causes, and consequences

The business investigation identifies canceled orders due to variant-stock discrepancies, size-related exchanges, incorrect color or size in packing, mismatches between displayed color and the product, and delayed consumption reports. The IDs below permit a direct link to needs and system requirements.

ID	Problem	Cause described in the business investigation
VD01	Orders canceled because variant stock is inaccurate.	Spreadsheet quantities use general product codes rather than timely color-size SKU updates.
VD02	Frequent size exchanges.	Insufficient standard size charts and sizing support.
VD03	Wrong color or size picked and packed.	Pick lists do not clearly identify variant SKUs.
VD04	Displayed colors do not match the actual goods.	No consistent color and photography standards.
VD05	Slow sales and consumption analysis.	Fragmented data and manual aggregation by color and size.

Table 1.6: Business problems and their stated causes.

VD01 undermines fulfillment and trust; VD02 raises exchange work and transport costs; VD03 results in wrong deliveries; VD04 changes customer expectations; and VD05 delays collection and inventory decisions. These are qualitative effects. The supplied material does not provide a dataset from which to calculate actual rates or demonstrate improvement.

1.5.2 Problem tree

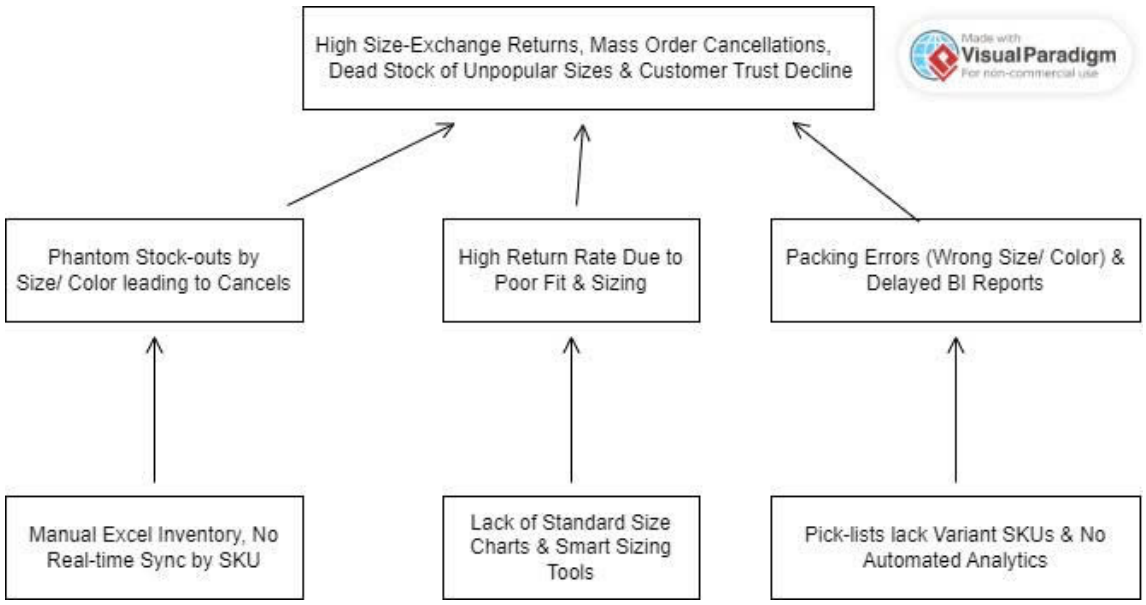


Figure 1.3: Problem tree: causes, intermediate problems, and shared consequences.

The left branch links spreadsheet inventory and missing real-time SKU synchronization to apparent size/color stock-outs and cancellations (VD01). The middle branch links weak

size guidance to poor fit and returns (VD02). The right branch links SKU-free pick lists and a lack of automated analysis to packing errors (VD03) and delayed reports (VD05). The common outcomes include size exchanges, cancellations, dead stock, and falling trust. VD04 is discussed in the business investigation text but is not drawn as a separate branch in this particular figure; it remains in the problem table for completeness.

1.6 Proposed Information System

1.6.1 Purpose, objectives, and scope

The system should unify fashion inventory by SKU, support customer selection, connect orders to warehouse work, and provide reliable management information. Its objectives are to limit orders for unavailable variants, assist sizing decisions where specified, improve picking accuracy, track delivery and returns, and consolidate sales and inventory information.

The functional scope follows FR01–FR20 of the requirements specification: accounts and profiles; categories, products and variants; text, voice and image search; product details; cart and checkout; payments and refunds; SKU pick lists, inventory and returns; shipping; comments; and reports. Priorities are addressed later in this chapter. An augmented-reality fitting room is outside the current scope. Advanced wish lists and outfit recommendations are optional extensions. AI examples, promotions, supplier ordering, and sizing advice appear in the background documents, but the requirements specification does not provide sufficiently specific functional requirements to treat them as commitments under FR01–FR20.

1.6.2 System boundary and context diagram

The software stores and processes business information. Payment services handle money movement; shipping services move parcels. Staff members remain external actors to the software boundary even though they belong to ABC Fashion as an organization.

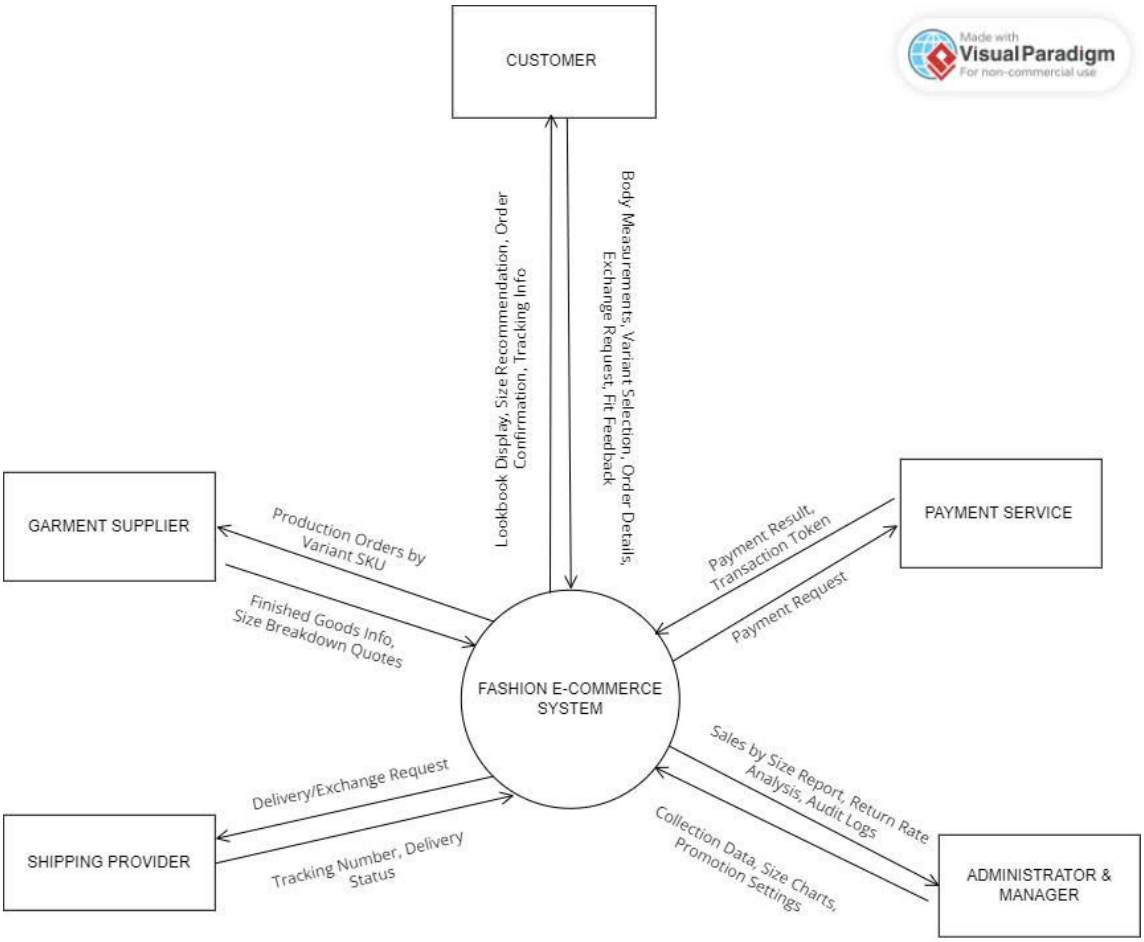


Figure 1.4: Preliminary context diagram with five external parties.

The customer sends orders, measurements, exchange requests, and feedback, and receives catalog information, confirmations, and tracking details. Administrators and managers provide collection data, size charts, and settings, and receive sales and returns information. The payment service exchanges requests, responses, and transaction references. The shipping provider receives delivery/exchange requests and sends waybill numbers and statuses. The garment supplier appears in this *preliminary* diagram with production orders and finished-goods information. The requirement catalogue has no corresponding direct FR or Use Case, so supplier integration would require a separate detailed specification. Sales, service, and marketing staff are Chapter 1 stakeholders even though the Use Case diagram does not draw separate cases for them.

1.6.3 Detailed external information flows

- Customer → system: search input, body measurements, color/size choice, cart and order information, exchange request, and fit feedback. System → customer: variant images and availability, possible sizing advice, confirmation, and tracking updates.
- Garment supplier → system: production progress, finished quantities by size and color, and quotations. System → supplier: proposed manufacturing orders by SKU and

target delivery dates. These preliminary context flows are not a committed supplier integration in FR01–FR20.

- System → payment service: order identifier and payable amount. Payment service → system: success/failure response, reference, and reconciliation information.
- System → shipping provider: delivery or exchange-pickup request, address, parcel information, and any COD amount. Shipping provider → system: waybill code and delivery/return status events.
- Administrator and manager → system: catalog and size-chart data, color standards, and report queries. System → them: collection sales, variant consumption, return analysis, and audit information where available.

1.6.4 Six business objectives

1. Synchronize inventory by color-size SKU. The investigation proposes reducing unavailable-stock orders by 90%; this is a target, not an observed result, and requires a baseline.
2. Offer measurement-based sizing assistance to reduce size-related returns. The concept appears in the business scope and AI illustration; its acceptance criteria need a separate FR.
3. Generate unambiguous SKU pick lists and support barcode checking to reduce wrong-color and wrong-size fulfillment.
4. Standardize digital color codes and lookbook photography; display software alone cannot guarantee the appearance of physical goods on all devices.
5. Coordinate size-exchange requests and reverse delivery with the shipping provider. Return conditions still require agreement.
6. Analyze sales by collection, color, and size to plan replenishment and reduce stagnant stock.

1.7 Functional Requirements

1.7.1 Organization and source of the requirements

Functional requirements describe what the system must do, without prescribing its implementation. The next two tables define the shared FR01–FR20 catalogue. Here M means Must Have and S means Should Have. The stakeholder/need column is an analytical trace to Section 1.3, not a record of interviews.

ID	System behavior	Need or party	Priority
FR01	Allow customer registration and login.	Customer	M
FR02	Allow customers to manage profile information.	Customer	S
FR03	Allow administrators to manage categories and variant matrices.	NC04	M
FR04	Allow administrators to create, update, and delete products.	NC04	M
FR05	Allow product search using keywords, voice, and images.	NC01	M
FR06	Display product details, including variant information.	NC02	M
FR07	Add the selected product variant to the cart.	NC02	M
FR08	Display cart contents.	NC02	M
FR09	Check out the cart and place an order.	NC03	M
FR10	Generate a warehouse pick list by SKU after order placement.	NC06	M

Table 1.7: Functional requirements FR01–FR10.

FR01 combines registration and login because the source groups them; individual test cases may still separate the two behaviors. For FR05, accepted voice/image input, recognition failure, and response time require further definition. The full implementation of a search algorithm is a design question, not a requirement stated by these two chapters.

ID	System behavior	Need or party	Priority
FR11	Let customers pay for orders securely.	Customer	M
FR12	Process online payments through the integrated service.	NC09	M
FR13	Verify payment status and handle refunds.	NC09	M
FR14	Allow the shipping provider to create waybills.	NC10	M
FR15	Update delivery and exchange/return status.	NC03, NC10	S
FR16	Allow warehouse staff to view pick lists and update inventory by SKU.	NC06	M
FR17	Process returned items and update the appropriate stock state.	NC07	M
FR18	Allow customers to submit product comments and feedback.	NC08 (partial)	S
FR19	Let managers view analytics dashboards on sales, stock, and returns.	NC05	S
FR20	Generate comprehensive management reports.	NC05	S

Table 1.8: Functional requirements FR11–FR20.

1.7.2 Associated business rules

1. **Variant identity:** a cart or order line must identify a SKU and its color and size, not merely a product design.
2. **Availability:** an order must not be confirmed when a selected SKU cannot satisfy the requested quantity. Checking and reserving stock must be consistent under concurrent orders.
3. **Order total:** the amount derives from quantities and line prices; shipping, promotions, and adjustments still need precise rules.
4. **Payment state:** choosing cash on delivery (COD) does not mean that money has already been collected.
5. **Returns:** an incoming item is associated with its original SKU; inspection determines whether it can re-enter saleable stock.

The requirement catalogue does not settle when stock is reserved and deducted, how failed payments release reservations, or exactly how COD is represented. Chapter 2 therefore makes these open questions explicit rather than silently assigning a policy.

1.8 Non-Functional Requirements

Ten NFR identifiers are used consistently. The figures of two seconds, 99.9%, and 5,000 are proposed targets from the requirements, not measured results. Test conditions must be agreed before acceptance.

NFR01 – Performance
: Normal product searches by text, voice, or image should return in two seconds according to the requirements specification. The meaning of peak load, dataset size, system configuration, and timing boundary remains open.

NFR02 – Security
: Only authenticated and authorized users may access personal order information. Tests should include unauthenticated requests and attempts to read another customer's order.

NFR03 – Usability
: Customer pages should use consistent navigation and legible color/size labels. Color fidelity requires accepted photography and display conditions; software alone cannot guarantee identical rendering on every screen.

NFR04 – Availability
: Target availability is at least 99.9% during business hours. The operating window, measurement period, and treatment of maintenance must be defined.

NFR05 – Reliability

Confirmed orders and their stock effects must remain consistent and retrievable, including after an interrupted transaction.

NFR06 – Scalability

The requirement catalogue refers to 5,000 "active users and SKUs." Users and SKUs are different quantities; the concurrency level and variant count must be specified separately.

NFR07 – Maintainability

Changes to one functional area should be testable without uncontrolled effects on others. The modular architecture mentioned in the earlier specification is a later design decision.

NFR08 – Data integrity

Prevent confirmation of orders with invalid SKUs or insufficient stock; this constrains FR09 and FR16.

NFR09 – Compatibility

Support suitable desktop and mobile browsers; named versions and viewport sizes remain to be agreed.

NFR10 – Compliance

Handle customer data in accordance with the project's applicable commerce/privacy rules. The requirement catalogue does not identify the exact legal criteria, so compliance cannot yet be certified.

1.9 Requirement Priorities

1.9.1 Must Have

The requirements are marked FR01, FR03–FR14, FR16–FR17, NFR02, NFR05, and NFR08 as essential for catalog integrity, ordering, payment, warehouse work, and access control.

1.9.2 Should Have

FR02, FR15, FR18–FR20, NFR01, and NFR04 support profiles, tracking, feedback, reports, performance, and availability. The AI sizing discussion in the requirements specification does not establish a separate FR identifier in the latest list.

1.9.3 Could Have and Won't Have

An advanced wish list and outfit recommendations are potential extensions without detailed FR identifiers. An AR fitting room is excluded from the present release. The requirement list gives no explicit priority to NFR03, NFR06, NFR07, NFR09, or NFR10;

their priority must be confirmed, not inferred to be low. FR14 is Must Have while the broader tracking FR15 is Should Have: even the first release must still handle a failed waybill request honestly.

1.10 Traceability and Requirement Review

1.10.1 From problem to need to requirement

Problem	Need	Requirement	Contribution and limitation
VD01	NC04	FR03	Identifies color-size variants consistently.
VD01	NC06	FR16, NFR08	Maintains SKU stock and prevents impossible orders.
VD01	NC03	FR09	Checks stock again before order confirmation.
VD02	NC02	FR06	Displays details; specific size recommendations lack an FR.
VD03	NC06	FR10, FR16	Gives warehouse staff SKU-level picking information.
VD04	NC02	FR06, NFR03	Shows product images; photography standards remain separate.
VD05	NC05	FR19, FR20	Provides dashboards and consolidated reports.

Table 1.9: Representative problem–need–requirement links.

1.10.2 Clarity, completeness, and feasibility

FR03, FR05, FR07, FR10, and FR19 have identifiable inputs or outputs and can support concrete tests. That does not mean the specification is complete. VD04 is covered only in part by image-display and usability requirements. NC08 lacks a complete customer-service workflow; NC10 needs service error and data-exchange rules. Refund timing, stock release after payment failure, return eligibility, and report definitions remain unresolved.

The proposed NFR01 two-second target cannot be tested without a load profile; NFR04 requires a business-hours calendar and measurement period; NFR06 needs separate user and SKU scales. FR09 can be checked with valid, unavailable-stock, and payment-failure scenarios. The supplied documents do not establish that performance targets are already feasible or met.

Chapter 2

REQUIREMENTS ANALYSIS

This chapter develops the requirements established in Chapter 1 into system actors, Use Case descriptions, Activity Diagrams, Swimlane Diagrams, requirement traceability, and validation. The identified functions follow the FR01–FR20 catalogue.

2.1 Actor Analysis

An actor represents an external role that directly interacts with the proposed system. The six actors in the ABC Fashion Use Case diagram are identified below.

1. **Customer:** Interacts through the web or mobile interface to register, log in, search by keyword, voice or image, view products, add variants to the cart, check out, pay, and submit comments or feedback.
2. **Administrator:** Manages product categories, products, and variant information.
3. **Warehouse Staff:** Views pick-lists, picks and packs items by SKU, updates inventory, and processes returned items.
4. **Manager:** Accesses analytics dashboards, sales reports, inventory reports, and return reports.
5. **Payment Service:** External service that processes online payments, verifies payment status, and executes refunds.
6. **Shipping Provider:** External service that creates waybills and updates delivery and exchange/return statuses.

The garment supplier is a stakeholder in the preliminary context diagram from Chapter 1. However, it is not drawn as a direct actor in the supplied Use Case diagram.

2.2 Use Case Model

2.2.1 Actor Identification

The six actors are identified from the system boundary and external service interactions: Customer, Administrator, Warehouse Staff, Manager, Payment Service, and Shipping Provider. Staff members are actors because they interact with the software, even though they belong to ABC Fashion as an organization.

2.2.2 Use Case Identification

The Use Case diagram contains functions for account access, product browsing and search, shopping-cart operations, orders, payments, product and inventory management, reporting, and shipping integration. Five key use cases are described in detail in Section 2.3.

ID	**Use Case**	**Primary actor(s)**	**Related FRs**
UC01	Search Products	Customer	FR05
UC02	View Products	Customer	FR06
UC03	Place Order / Checkout	Customer	FR07–FR09, FR11–FR14
UC04	Manage Products & Inventory	Administrator, Warehouse Staff	FR03–FR04, FR10, FR16–FR17
UC05	View Analytics Dashboard & Reports	Manager	FR19–FR20

Table 2.1: The five principal Use Cases and their corresponding requirements.

2.2.3 UML Use Case Diagram

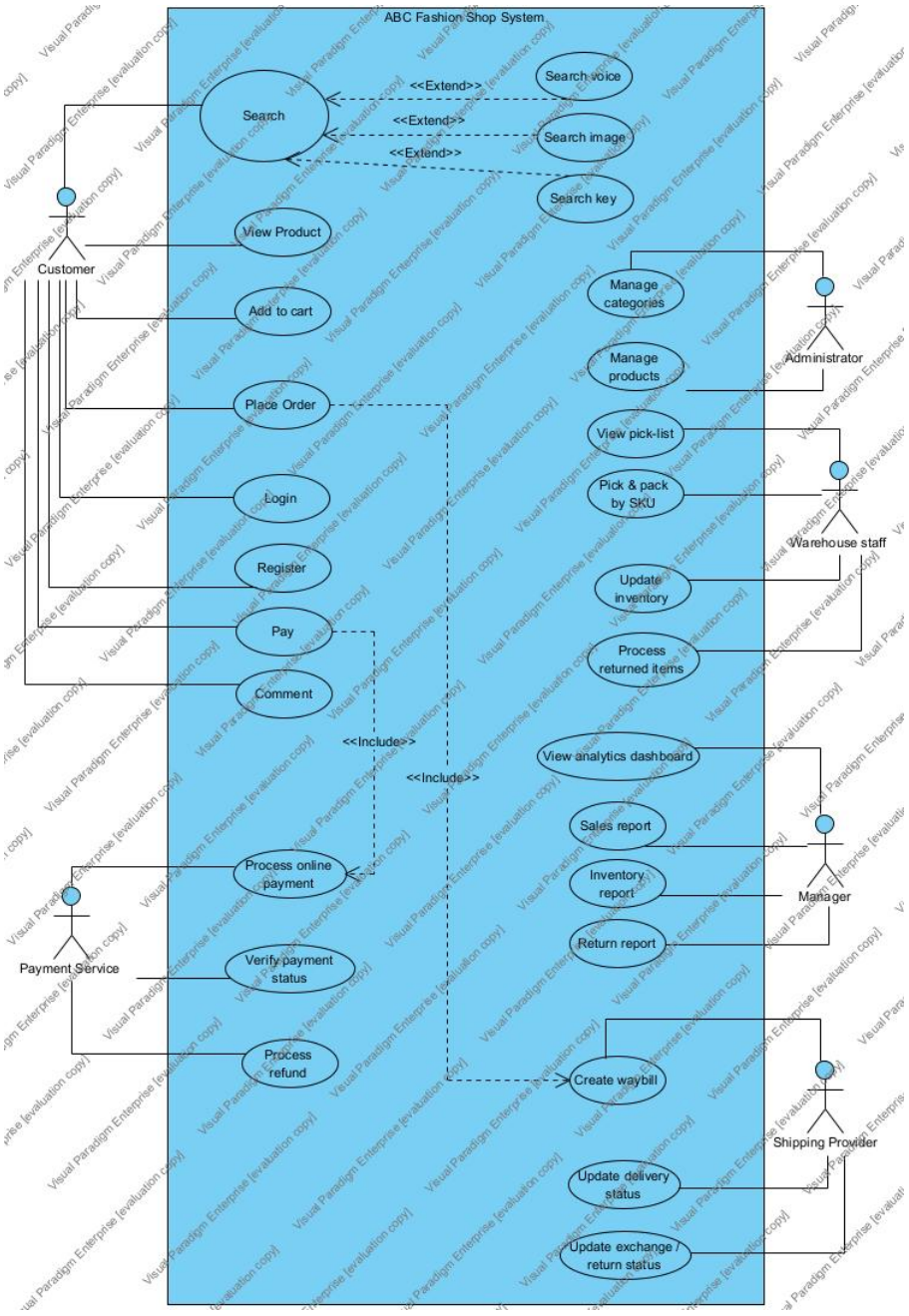


Figure 2.1: UML Use Case Diagram of ABC Fashion Shop System.

The diagram associates the Customer with search, product viewing, cart, ordering, login, registration, payment and comments; the Administrator with category and product management; Warehouse Staff with pick-lists, packing, inventory and returned items; the Manager with dashboards and reports; and the external services with payment and shipping functions.

2.2.4 Include and Extend Relationships

The original model identifies these relations:

- **Include:** *Place Order* and *Create Waybill*; *Pay* and *Process Online Payment*.
- **Extend:** *Search* and the three search forms: *Search Key*, *Search Voice*, and *Search Image*.

These relations are described as shown in the supplied diagram. The three search methods are alternatives for a user request; online payment is conditional on the chosen payment method, while COD does not require immediate online payment.

2.3 Use Case Specifications

The following five specifications retain the goal, actors, trigger, preconditions, postconditions, main scenario, alternative flows, exceptions, rules, and priority of the original Assignment 02 Use Cases.

2.3.1 Use Case 1: Search Products

Field	Specification
Use Case ID	UC01
Use Name	Search Products
Primary Actor	Customer
Goal	Find specific fashion products quickly using available search methods.
Trigger	The customer uses the search bar or selects a search option on the storefront.
Preconditions	The system is online and the product catalogue is loaded.
Postconditions	Matching products or variants are displayed, or the customer exits the search flow.
Main Success Scenario	1. Customer initiates a search. 2. System provides the search interface. 3. Customer chooses keyword, voice, or image input. 4. System queries the product catalogue. 5. System displays matching products.
Alternative Flows	**3a. Voice/Image:** The customer submits a voice query or image; the system interprets it into product attributes before querying the catalogue. **5a. Search Again:** The customer changes the query and repeats the search.
Exceptions	**No Results Found:** The system displays a no-results message and alternative suggestions; the customer can search again or exit.

Field	Specification (continued)
Business Rules	Results display active, stocked products or products explicitly available for pre-order.
Priority	Must Have

Table 2.2: UC01 – Search Products specification.

2.3.2 Use Case 2: View Products

Field	Specification
Use Case ID	UC02
Use Name	View Products
Primary Actor	Customer
Goal	View product details, lookbook images, and available color/size variants.
Trigger	The customer selects a product from a search result or collection catalogue.
Preconditions	The system is online and product data is loaded.
Postconditions	The product-detail page shows the available variants and prices.
Main Success Scenario	1. Customer selects a product. 2. System retrieves product details. 3. System checks availability. 4. System displays the product name, description, category, price, lookbook images, color swatches, size matrix, and real-time stock. 5. The product-detail page is displayed.
Alternative Flows	**5a. Color Variant Selected:** The customer selects a color swatch; the system updates the lookbook images, price, and variant stock displayed.
Exceptions	**Product Unavailable:** The system shows an unavailable message and redirects the customer to the catalogue.
Business Rules	Product information must reflect the maintained variant attributes and prices.
Priority	Must Have

Table 2.3: UC02 – View Products specification.

2.3.3 Use Case 3: Place Order / Checkout

Field	Specification
Use Case ID	UC03
Use Name	Place Order / Checkout
Primary Actor	Customer
Supporting Actors	Payment Service, Shipping Provider
Goal	Successfully purchase selected items from the cart.
Trigger	Customer chooses Checkout from the shopping cart.
Preconditions	Customer is logged in and the shopping cart contains at least one item.
Postconditions	An accepted order has corresponding inventory and payment states; a waybill is recorded when creation succeeds.
Main Success Scenario	1. Customer initiates checkout. 2. System checks the cart and real-time SKU stock. 3. System calculates costs and shipping fees. 4. Customer selects the payment method. 5. For an online method, the Payment Service processes the payment. 6. System creates the accepted order, updates inventory by SKU, and requests a waybill from the Shipping Provider. 7. System displays order confirmation.
Alternative Flows	**2a. Invalid Cart / Out of Stock:** The system asks the customer to correct the cart. **4a. COD:** Online collection is skipped and the order proceeds with payment awaiting collection.
Exceptions	**5a. Payment Failure:** System shows the failure, does not confirm the order, and preserves the cart. If the shipping provider fails after an order is accepted, its waybill remains pending rather than showing a nonexistent tracking number.
Business Rules	Inventory is checked at the SKU level before confirming the order; reservation and deduction must not be counted twice.
Priority	Must Have

Table 2.4: UC03 – Place Order / Checkout specification.

2.3.4 Use Case 4: Manage Products & Inventory

Field	Specification
Use Case ID	UC04
Use Name	Manage Products & Inventory
Primary Actors	Administrator, Warehouse Staff
Goal	Maintain accurate catalogue information, SKU-level stock, and picking and return records.
Trigger	Administrator opens product management, or Warehouse Staff receives a pick-list.
Preconditions	Staff members have appropriate authentication and role permissions.
Postconditions	Product information and warehouse stock records reflect the processed operations.
Main Success Scenario	1. Warehouse staff opens an assigned pick-list. 2. Staff picks and packs items by SKU. 3. System updates stock by SKU. 4. System checks concurrent stock changes. 5. For received returns, staff inspects the items and the system records resalable stock or damage. Separately, administrators maintain the product catalogue.
Alternative Flows	**5a. No Return Received:** The fulfillment flow ends without restocking. **5b. Defective Return:** The item is recorded as defective instead of available stock.
Exceptions	**Concurrency Conflict:** Concurrent updates to one SKU must be logged and handled under an agreed stock-consistency rule. The source scenario describes a latest-update policy, which requires review because it can lose stock changes.
Business Rules	Inventory movements must refer to the actual color/size variant SKU.
Priority	Must Have

Table 2.5: UC04 – Manage Products and Inventory specification.

2.3.5 Use Case 5: View Analytics Dashboard & Reports

Field	Specification
Use Case ID	UC05

Field	Specification (continued)
Use Name	View Analytics & Generate Reports
Primary Actor	Manager
Goal	Monitor business performance through dashboards and management reports.
Trigger	The manager opens the analytics section of the management portal.
Preconditions	The manager is authenticated with managerial privileges.
Postconditions	Sales, stock and return reports are compiled and displayed.
Main Success Scenario	1. Manager selects a report. 2. System checks whether a custom date range is supplied. 3. System queries completed transactions and relevant inventory/return data. 4. System checks export volume. 5. System compiles charts and exportable tables and displays the report.
Alternative Flows	**2a. Custom Date Range:** The selected period is applied before querying. **2b. Default Timeframe:** The system uses the default report period when no custom range is supplied.
Exceptions	**4a. Massive Export Limit:** The system limits batch size and offers CSV download rather than rendering an excessively large dataset.
Business Rules	Financial reporting must use an agreed definition of successfully completed transactions.
Priority	Should Have

Table 2.6: UC05 – Analytics and Reports specification.

2.4 Activity Diagrams

Activity Diagrams illustrate the order of work, branching conditions and any parallel processing for five core workflows from the source report.

2.4.1 Activity Diagram 1: Search Products Flow

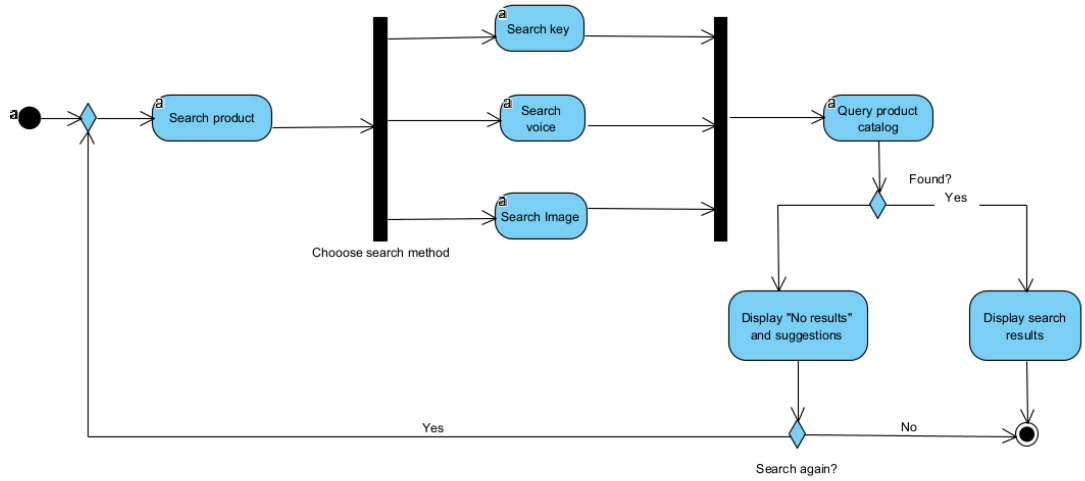


Figure 2.2: Activity Diagram: Search Products Flow.

The customer starts a search, selects a keyword, voice or image method, and the system queries the catalogue. The diagram uses parallel fork/join bars across the three search inputs; logically these represent alternative input choices for a search request. When matches exist, the system displays search results. Otherwise, it shows no results and suggestions, allowing the customer to search again or exit.

2.4.2 Activity Diagram 2: Add to Cart Flow

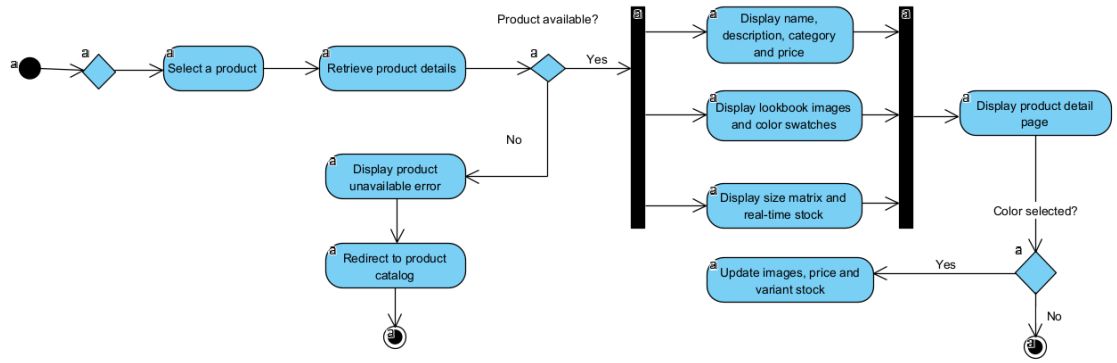


Figure 2.3: Activity Diagram: Add to Cart Flow.

The Add to Cart flow starts with product selection and an availability check. Unavailable products return to the catalog with an error message. Otherwise, the system displays product details, lookbook images, color swatches, sizes, prices, and stock. Choosing a color updates the corresponding images, price, and variant availability, helping the customer choose the correct item before adding it to the cart (FR07).

2.4.3 Activity Diagram 3: Place Order / Checkout Flow

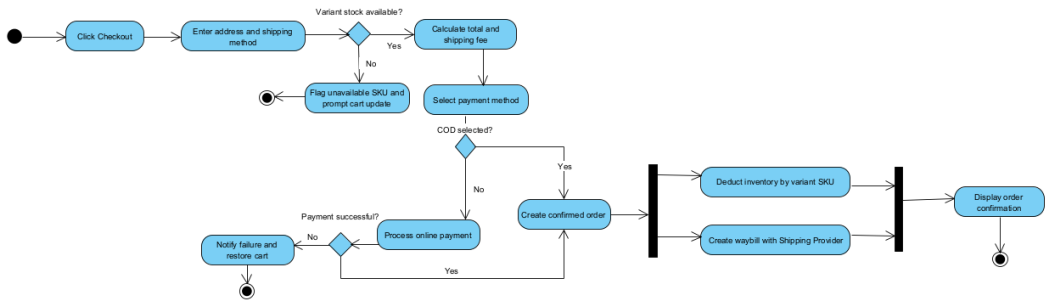


Figure 2.4: Activity Diagram: Place Order and Checkout Flow.

The customer opens checkout, supplies the shipping address, and the system validates the order and calculates the payable amount. The customer chooses a payment method. Online payment proceeds through the Payment Service; COD bypasses immediate online collection. A valid order triggers stock processing and waybill generation. An unsuccessful payment does not produce a successful-order confirmation.

2.4.4 Activity Diagram 4: Warehouse Fulfillment Flow

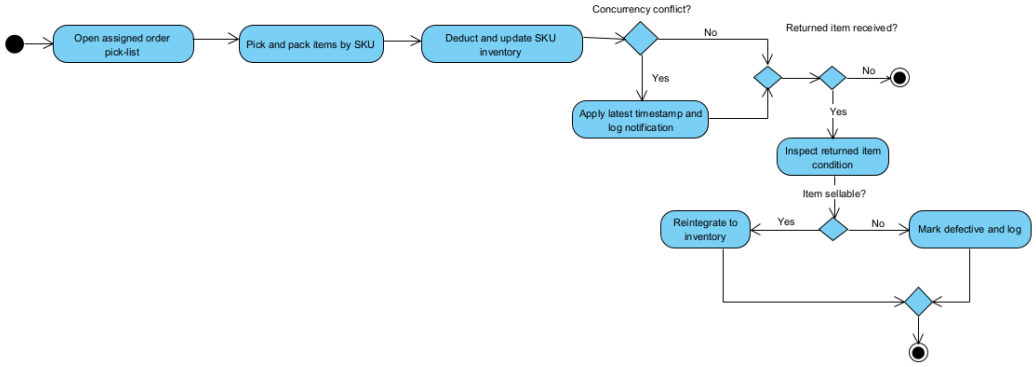


Figure 2.5: Activity Diagram: Warehouse Fulfillment Flow.

Warehouse Staff opens a generated order pick-list, picks and packs items according to SKU and shelf location, and the system updates stock. The diagram then checks whether returned items have been received. If so, they are inspected; accepted stock is returned to inventory, while defective items are marked as defective. The final inventory state must distinguish saleable returns from damaged stock.

2.4.5 Activity Diagram 5: Reporting and Analytics Flow

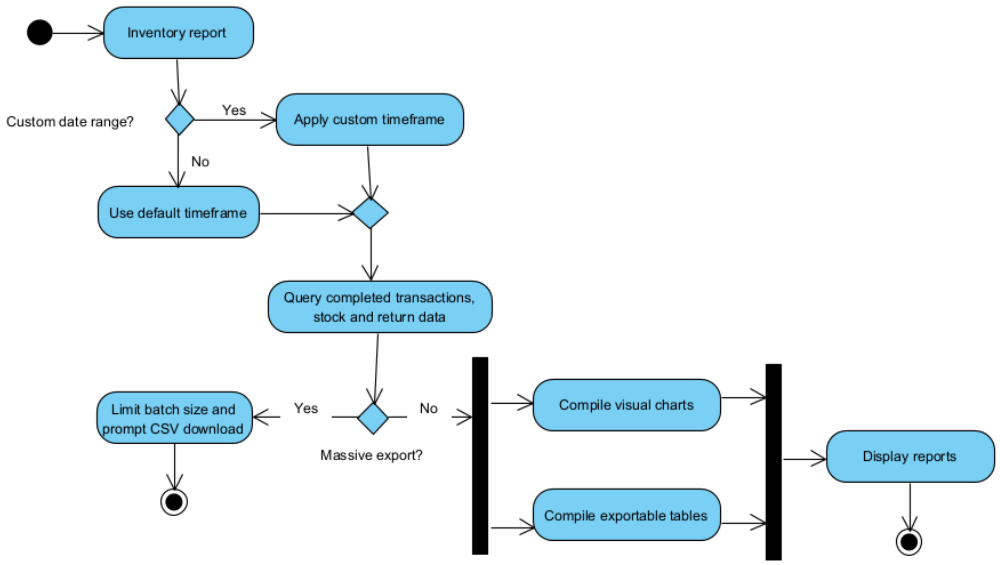


Figure 2.6: Activity Diagram: Reporting and Analytics Flow.

A manager chooses an inventory or analytics report and may supply a custom period. The system queries completed transactions, stock and return data, checks whether the export is too large, and compiles visual charts and exportable tables. An excessive export leads to a limited-batch or CSV-download branch. The figure uses parallel work for chart and table creation, unlike the alternative methods in the search flow.

2.5 Swimlane Diagrams

Swimlane Diagrams show how activities are distributed among organizational roles and external services.

2.5.1 Swimlane Diagram 1: Order Processing & Payment Workflow

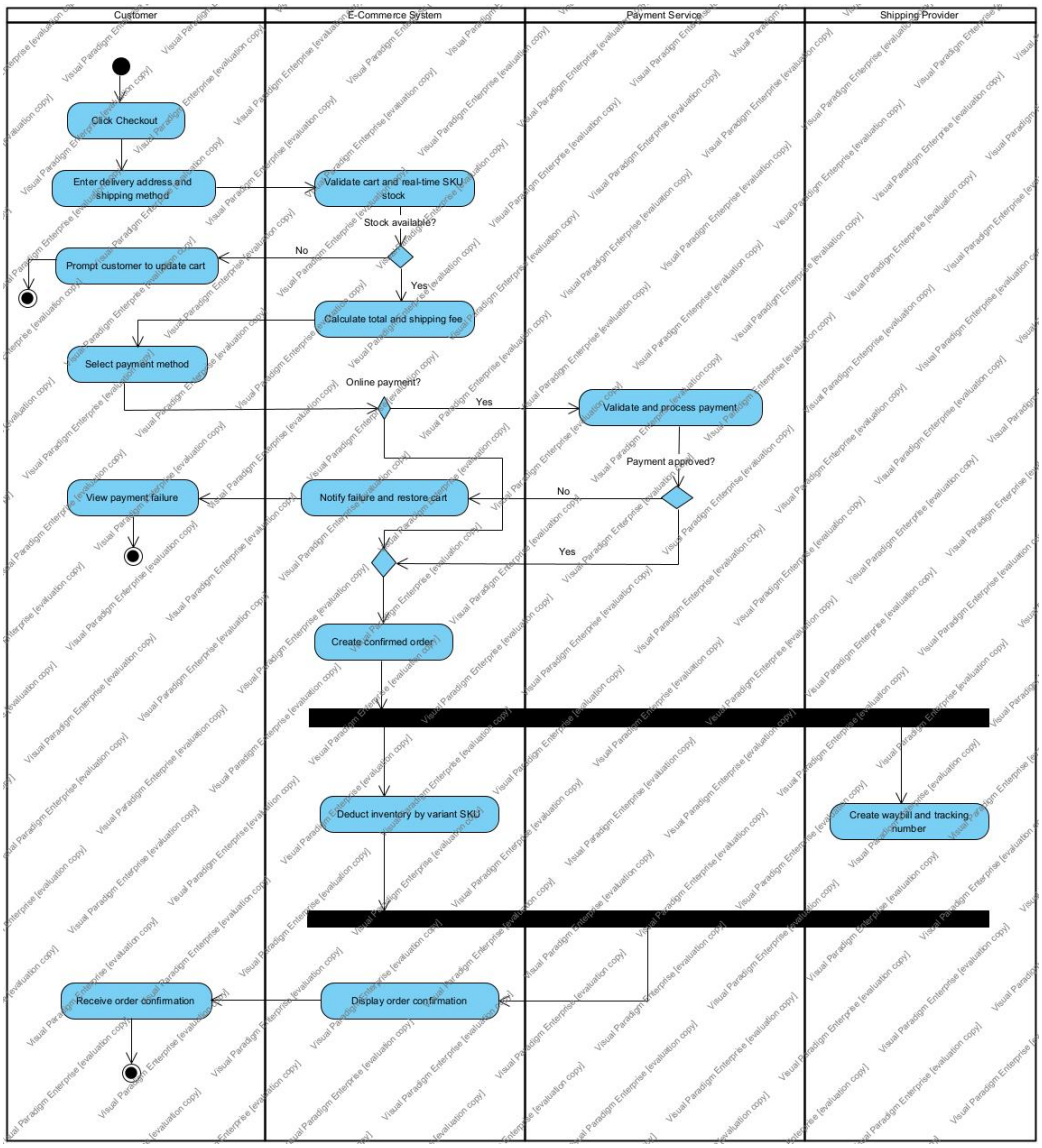


Figure 2.7: Swimlane Diagram: Order Processing and Payment Workflow.

The diagram has lanes for the Customer, E-Commerce System, Payment Service, and Shipping Provider. The customer starts checkout and provides address and payment information. The system validates stock and calculates costs; the Payment Service verifies the selected online payment when applicable. The system generates the order, coordinates SKU-level inventory handling, and requests a waybill from the Shipping Provider. The customer receives the order confirmation once the corresponding transaction state is known.

2.5.2 Swimlane Diagram 2: Inventory Receiving Workflow

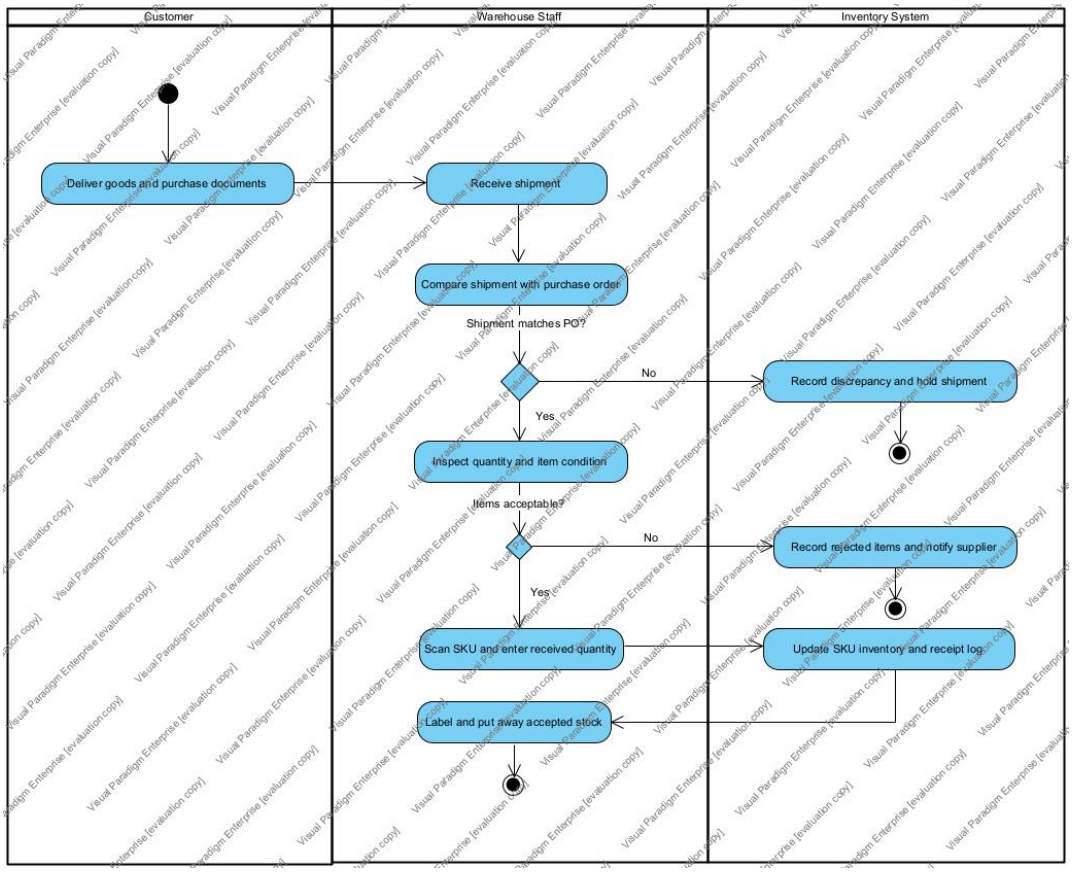


Figure 2.8: Swimlane Diagram: Inventory Receiving Workflow.

The source text describes the Supplier delivering goods and purchase documents; Warehouse Staff receives the shipment, checks the purchase order, and inspects quantities and conditions; the Inventory System updates accepted stock or records discrepancies. The image itself names its first lane "Customer", despite showing a supplier-receiving process. This inconsistency is present in the supplied graphic. Inventory receiving from a supplier is distinct from the customer-return procedure defined by FR17.

2.6 Requirement Traceability

Traceability links the main business problems identified in Chapter 1 to stakeholder needs, requirements, and Use Cases.

Business problem	Stakeholder need	Requirement	Use Case
Inaccurate variant inventory causing cancelled orders	Accurate stock down to SKU	FR16: Update inventory by SKU	UC04: Manage Products & Inventory
High size-exchange returns due to poor fit	Clear product attributes and size information	FR06: View product details	UC02: View Products
Packing errors: wrong color or size shipped	Pick-lists showing variant SKUs	FR10: Generate pick-lists by SKU	UC04: Manage Products & Inventory
Delayed analytics on sales trends	Timely consumption reports	FR19: View analytics dashboards	UC05: View Analytics Dashboard & Reports

Table 2.7: Business problem–stakeholder need–requirement–Use Case traceability.

The links above reflect the four traceability examples in the source A02 report. They are representative rather than an exhaustive mapping of every FR01–FR20 entry.

2.7 Requirement Validation

Requirement validation evaluates whether the requirements are clear, consistent, complete, and testable against business rules. FR05, FR10, FR16, NFR01 and NFR08 are suitable starting points for verification, subject to defined test conditions.

Problematic wording	Revised requirement wording
"The system should be fast."	**NFR01:** The system shall return normal product-search results (keyword, voice, image) within 2 seconds under peak operating conditions. Peak load and test data still need to be defined.
"The system should be user-friendly."	**NFR03:** The system shall provide consistent navigation menus across customer-facing product pages. A checklist must define what consistency means.
"The system must protect user data."	**NFR02:** Only authenticated and authorized users shall access their personal order and payment details via secured authentication mechanisms. Verification must include access-control tests.

Table 2.8: Three examples of requirement clarification and validation.

References

[1] Dinh Que Tran (2026). *Information System Analysis and Design – Assignment 01: E-Commerce Information System Investigation*. Department of Information Technology, PTIT.

[2] Dinh Que Tran (2026). *Information System Analysis and Design – Assignment 02: E-Commerce Requirements Engineering and Specification*. Department of Information Technology, PTIT.